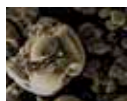




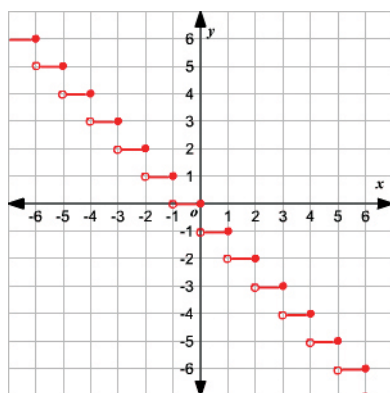
UBER to pay pension

Uber is to pay millions of pounds in missed pension payments to UK drivers.
<https://digit.in/oct21-15>



Toxic metal eating bacteria

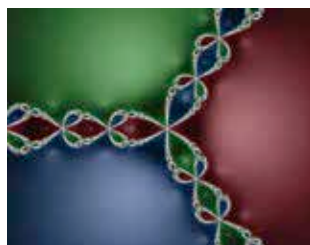
Scientists at the IIT-BHU have discovered a bacteria capable of separating toxic metal from water <https://digit.in/oct21-16>



Step function

You can define a step function as a function that steps abruptly from one constant value to another. This is also used in computer programming for various purposes and is widely used in cloud technologies. This is also known as a staircase function as it forms a structure like a staircase when plotted.

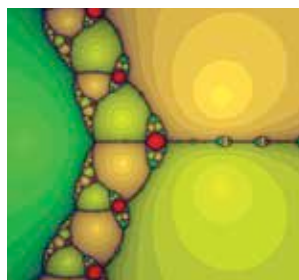
Newton Fractal for $f(z)=z^3-1$



The Newton fractal is a boundary set in the complex plane which is characterized by Newton's method applied to a fixed polynomial $p(Z) \in \{\mathbb{C}\}[Z]$ or transcendental function (wikipedia)

Newton Fractal for $f(z)=z^3-2z+2$

Another example of Newton's Fractal, at $f(z)=z^3-2z+2$ it gives a beautiful visualization that matches the microscopic images of roots of plants!



Menger sponge



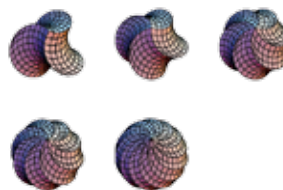
The Menger Sponge has a fractional dimension (technically referred to as the Hausdorff dimension) between a plane and a solid, approximately 2.73, and it has been used to visualize certain models of a foam-like space-time.

Mandelbulb

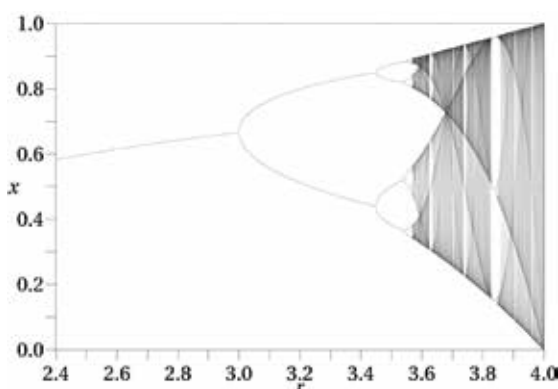
Yes, it is related to the Mandelbrot set. It is actually the 3D version of the Mandelbrot set but with spherical co-ordinates constructed for the first time in 1997 by Jules Ruis



Rhombic spiralohedron



The fun fact about this is that it was discovered accidentally by Russell Towle while attempting to develop a function to create a rhombic hexahedron from a triple of vectors. The result of the mistake looked something like this



Logistic Maps

Logistic is a mapping technique that uses polynomials of degree 2. The map was popularized in a 1976 paper by the biologist Robert May. This equation shows how even a simple, non-linear equation can show complex and chaotic behavior.

Polar equations

Polar equations are basically the equations to define the relationship between a radius and theta of a circle.

